

Quan Tran Hong

📍 Vietnam | ✉️ quanth238@gmail.com | 🌐 quanth238.github.io | 🌐 [quan2381](#)

Research Interests

3D vision, Generative Modeling, Optimization, Computer Vision, Neural Rendering, Probabilistic Machine Learning.

Education

VinUniversity

Master of Science in Computer Science

Graduate Research Excellence Program (full-tuition scholarship and stipend)

Hanoi, Vietnam

Mar 2026 – Present

Auckland University of Technology

Bachelor of Computer and Information Sciences

High Merit Graduate, GPA: 3.48/4.0 ([Transcript](#))

Sep 2019 – Aug 2023

Vietnam

Research Experience

Research Assistant, Binh AI Lab, CAIR, VinUniversity (Advisor: Asst. Prof. Binh T. Nguyen)

Mar 2026 – Present

- Research generative modeling, optimization, and probabilistic ML for 3D vision.
- Study training behavior of learned 3D representations for neural rendering and reconstruction.

Research Assistant, Cognitive Robotics Lab, University of Arkansas (Advisor: Dr. Tuan Dang)

Dec 2024 – Mar 2026

- Proposed and led a first-author project on geometric regularization for 3D Gaussian Splatting.
- Designed the method, ran experiments and ablations, and wrote the manuscript.
- Resulted in TriaGS, accepted to WACV 2026.

Publications

- [1] **Q. T. Hong**, L. Nguyen-Chi, and B. Nguyen. "Birth-Death Structural Learning for 3D Gaussian Splatting." *Submitted to the Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- [2] **Q. T. Hong**, L. Nguyen-Chi, and B. Nguyen. "Applying Splat Regression Models to Particle Density Control in Radiance Fields." *Submitted to the ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM)*.
- [3] **Q. T. Hong** and T. Dang. "TriaGS: Differentiable Triangulation-Guided Geometric Consistency for 3D Gaussian Splatting." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. [arXiv:2512.06269](#) / [code](#)
- [4] **Q. T. Hong** and B. Nguyen. "Render-Aware Subset Selection for Post-Training 3D Gaussian Splatting Compaction." Manuscript.

Talks

3D Gaussian Splatting (3DGS), invited talk, CAIR Monthly Reading Group.

Presented core concepts, training/rendering pipeline, and tradeoffs of 3DGS.

Mar 19, 2026

CAIR, VinUniversity

Service

Workshop Reviewer: ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM).

Teaching Experience

Teaching Assistant, Data Mining (COMP4040)

Prepared materials for Labs, led lab sessions, and supported student projects.

Feb 9, 2026 – May 2026

VinUniversity

Technical Skills

Languages: Python; JavaScript/TypeScript; familiar with Go, Java, and C++.

ML/CV: PyTorch, TensorFlow, Scikit-learn; training, fine-tuning, evaluation; 3D vision, generative models.

Systems: FastAPI, NestJS, PostgreSQL, Docker, AWS.

Industry Experience

Software Engineer , Fullerton Health Vietnam <i>Built healthcare backend services with microservices, SSO/RBAC, and optimized queries.</i>	Vietnam Dec 2023 – Oct 2025
Frontend Developer Intern , Zigvy Corporation <i>Built React UI components and implemented client-facing web application features.</i>	Vietnam Apr 2023 – Oct 2023

Competitions & Activities

- HCM AI Challenge 2024:** Advanced to the final round; built a web-based event-retrieval pipeline.
- AI Vietnam Program Graduate (Class of 2024):** Completed an intensive AI and data science training program.
- AWS Certified Cloud Practitioner:** Earned foundational AWS cloud certification.
- Full-Stack JavaScript Website Bootcamp:** Completed a six-month full-stack web development program.